

PROGRAMMING FUNDAMENTALS

Spring 2024

28th & 29th Feb

Ex-1

```
#include <iostream>
using namespace std;
int main()
{
    int b,x;
    b=10;
    x=2;
    b = x++;
    cout<<"Value in B = "<<b<<endl;
    cout<<"Now value in X = "<<x<<endl;

    b = ++x;
    cout<<"This time value in B is different"<<endl;
    cout<<"Value in B = "<<b;

    return 0;
}
```

Ex-1

```
#include <iostream>
using namespace std;
int main()
{
    int b,x;
    b=10;
    x=2;
    b = x++;
    cout<<"Value in B = "<<b<<endl;
    cout<<"Now value in X = "<<x<<endl;

    b = ++x;
    cout<<"This time value in B is different"<<endl;
    cout<<"Value in B = "<<b;

    return 0;
}
```

```
Value in B = 2
Now value in X = 3
This time value in B is different
Value in B = 4
-----
```

Switch Statement

- Instead of using the multiple **if** statements to code, you sometimes (but not always) can use the **switch statement**
- The switch statement begins with the switch clause
- Clause contains a *selectorExpression* enclosed in parentheses
- The *selectorExpression* must result in a value whose data type is bool, char, short, int, or long

Switch statement

- A statement block of one or more statements follows the switch clause enclosed in a set of braces
- You can include as many case clauses as necessary
- Each case clause represents a different alternative
- A case clause contains a value followed by a colon
- The data type of the value should be compatible with the data type of the *selectorExpression*
- When the *selectorExpression* is numeric → values in the case clauses should be numeric
- When the *selectorExpression* is a character → the values should be characters

Switch statement

- The expression following the word case must be an integer or constant
- It cannot be a variable
- It cannot be an expression such as $x < 22$ or $n == 50$

break statement

- The **break statement** tells the computer to leave (“break out of”) the switch statement at that point
- If you do not use it the computer continues processing the remaining instructions in the switch statement
- After processing the break statement, the computer processes the instruction that follows the switch statement’s closing brace

default clause

- The **default** clause can appear anywhere within the switch statement
- But is usually entered as the last clause in the statement
- At the **end, it does not need a break statement**
- You may include the break statement for clarity
- If the default clause is not the last clause, a break statement is required

Syntax

```
switch (selectorExpression)
{
  case value1:
    one or more statements
    [break;]
  [case value2:
    one or more statements
    [break;]]
  [case valueN:
    one or more statements
    [break;]]
  [default:
    one or more statements to be processed when the selector-  
Expression does not match any of the values in the case clauses
    [break;]]
}
```

//end switch

Example

```
int main()
{
    char grade = ' ';

    //enter grade
    cout << "Letter grade: ";
    cin >> grade;
    grade = toupper(grade);

    switch (grade)
    {
        case 'A':
            cout << "Excellent";
            break;
        case 'B':
            cout << "Above Average";
            break;
        case 'C':
            cout << "Average";
            break;
        case 'D':
        case 'F':
            cout << "Below Average";
            break;
        default:
            cout << "Invalid grade";
    } //end switch

    cout << endl;
    system("pause");
    return 0;
} //end of main function
```

Ex-2

- Write a program that accepts an integer value from 1 to 7 and displays the name of the day in a week using switch statement.

1 → Monday

2 → Tuesday

.

.

.

7 → Sunday

Ex-3

Lets assume that your program is meant to display factorial of any number from 2 to 10. Ask the user to enter an integer value. Using Switch case structure, display factorial of each value from 2 to 10 according to table:

2	2
3	6
4	24
5	120
6	720
7	5040
8	40320
9	362880
10	328800

 D:\Books-ICT\C-Programs\factorial_using_Switch.exe

Enter a number : 5

120

Process exited after 4.439 seconds with return

Press any key to continue . . .

 D:\Books-ICT\C-Programs\factorial_using_Switch.exe

Enter a number : 12

Invalid Input

Process exited after 3.337 seconds with return

Press any key to continue . . .

Ex-4

- Write C++ code that accepts the employee category and assign a bonus percentage. Then it also asks the user to enter the monthly salary and calculates the bonus amount.
- Category A earns 20% bonus, category B earns 15% bonus, category C earns 10% bonus.
- The code displays the salary, employee category and bonus amount and final salary earned.

Ex-4 - Sample Run

```
Enter Salary of the Employee = 100000
Enter the Category of the Employee (A,B,C) : A

Employee Category =  A
Bonus Amount = 20000
Final Salary Earned = 120000
-----
```

```
Enter Salary of the Employee = 110000
Enter the Category of the Employee (A,B,C) : R

Invalid Category Entered...
Run the program again....
-----
```

Using Switch in Menu Systems

- Very useful in designing Menus in your code
- Lets design a code that displays the Winner and Runner-up for the

Ex-5

- Design a menu for a restaurant where user can select any menu item. Then user can enter the quantity to order. The C++ code will then cost of every menu item and the total bill
- The user may also get some discount based on the bill amount as per following
- If the bill is up to 5,000 then you get no discount
- On a bill from 5001 to 10,000, you get 12% discount
- Otherwise, discount is 15%

Ex-6

Year	Winner	Runner-Up
2019	Quetta Gladiators	Peshawar Zalmi
2020	Karachi Kings	Lahore Qalandars
2021	Multan Sultans	Peshawar Zalmi
2022	Lahore Qalandars	Multan Sultans
2023	Lahore Qalandars	Multan Sultans

Write C++ code using Switch case structure to display the winner and the runner-up based on the entered year (integer value)

Book Reference

- D.S. Malik (ch - 4)
- Tony Gaddis (ch - 4)